

Nunit and MOQ

Name: DURGASREE AVVARU

Registered Mail: 2300031591cseh@gmail.com

Exercise 1:

Main.java:

```
© Main.java × pom.xml (JUnitDemo) © Calculator.java
1 package org.example;
2
3 public class Main {
4
5     public static void main(String[] args) {
6
7         System.out.println("Hello and welcome!");
8
9         for (int i = 1; i <= 5; i++) {
10             System.out.println("i = " + i);
11         }
12     }
13 }
```

Pom.xml:

```
© Main.java × pom.xml (JUnitDemo) × Calculator.java CalculatorTest.java
1 <project xmlns="http://maven.apache.org/POM/4.0.0"
2     xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3     xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
4         https://maven.apache.org/xsd/maven-4.0.0.xsd">
5
6     <modelVersion>4.0.0</modelVersion>
7
8     <groupId>org.example</groupId>
9     <artifactId>JUnitDemo</artifactId>
10    <version>1.0-SNAPSHOT</version>
11
12
13    <dependencies>
14        <dependency>
15            <groupId>junit</groupId>
16            <artifactId>junit</artifactId>
17            <version>4.13.2</version>
18            <scope>test</scope>
19        </dependency>
20    </dependencies>
21
22 </project>
```

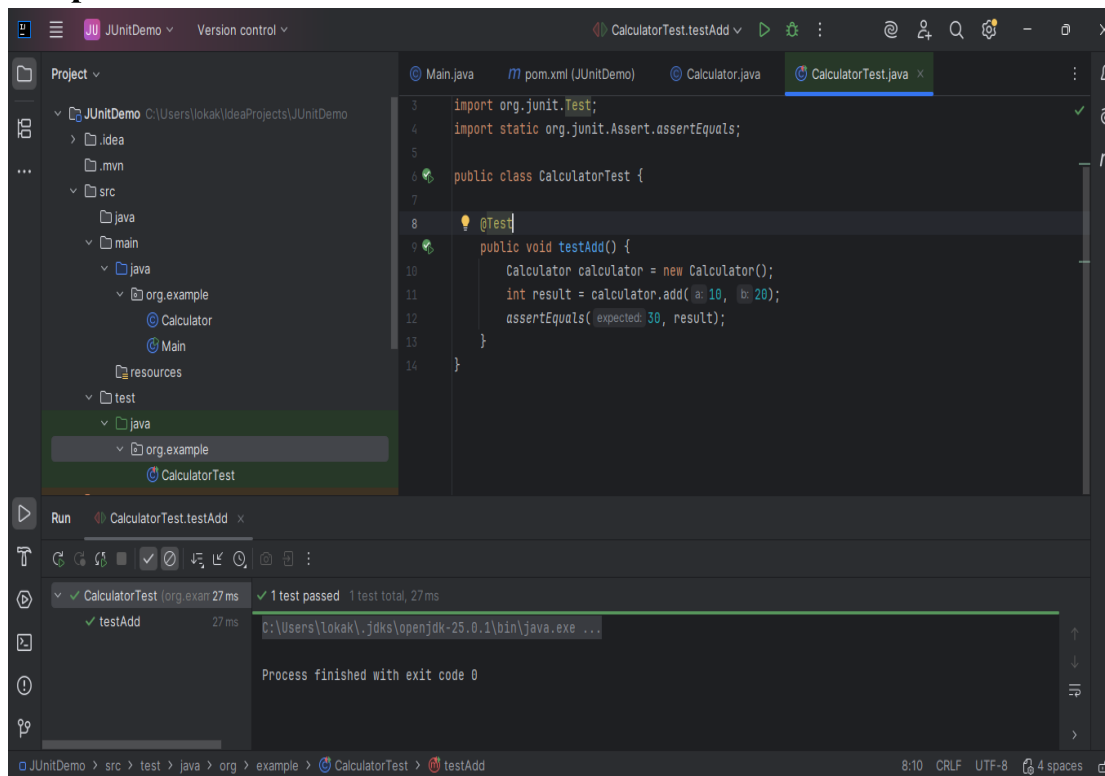
Calculator.java:

```
© Main.java    m pom.xml (JUnitDemo)    © Calculator.java ×
1      package org.example;
2
3      public class Calculator {  no usages
4
5          public int add(int a, int b) {  no usages
6              return a + b;
7          }
8      }
```

CalculatorTest.java:

```
© Main.java    m pom.xml (JUnitDemo)    © Calculator.java    © CalculatorTest.java ×
1      package org.example;
2
3      import org.junit.Test;
4      import static org.junit.Assert.assertEquals;
5
6      public class CalculatorTest {
7
8          @Test
9          public void testAdd() {
10             Calculator calculator = new Calculator();
11             int result = calculator.add(a: 10, b: 20);
12             assertEquals(expected: 30, result);
13         }
14     }
```

Output Screenshot:



Exercise 3: Assertions in JUnit:

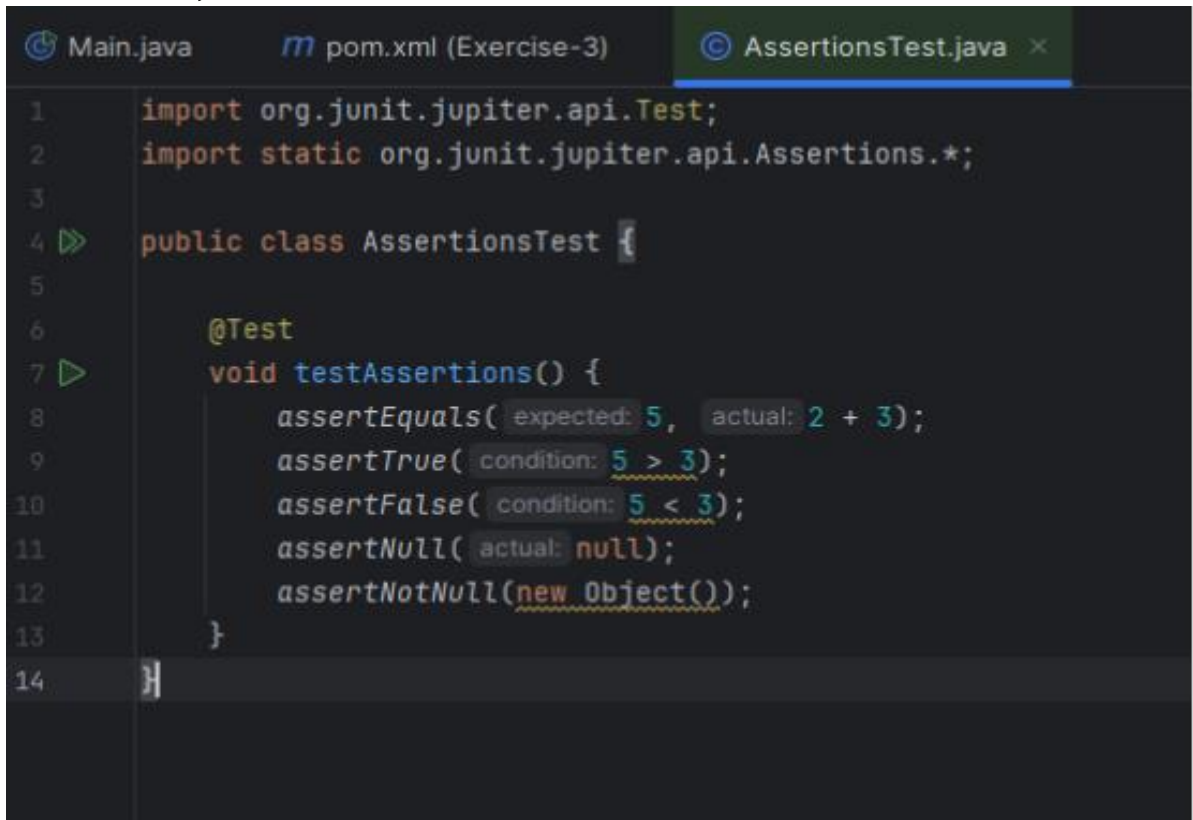
Main.java:

```
Main.java x pom.xml (Exercise-3) AssertionsTest.java
1 package org.example;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         System.out.println("Hello and welcome!");
7
8         for (int i = 1; i <= 5; i++) {
9             System.out.println("i = " + i);
10        }
11    }
12 }
```

Pom.xml:

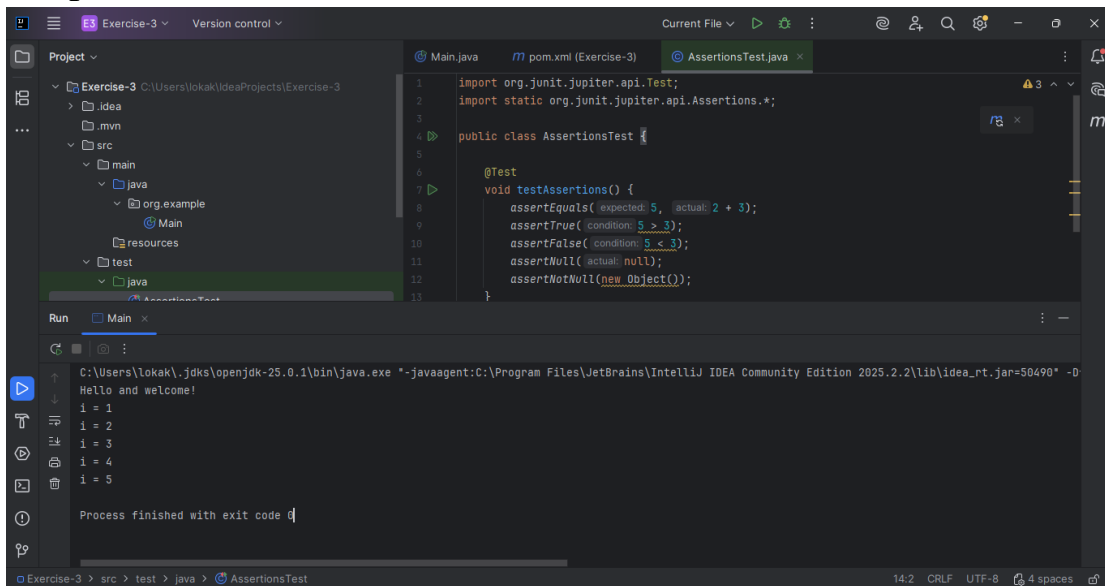
```
Main.java pom.xml (Exercise-3) x AssertionsTest.java
1 <?xml version="1.0" encoding="UTF-8"?>
2 <project xmlns="http://maven.apache.org/POM/4.0.0"
3         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
4         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
5     <modelVersion>4.0.0</modelVersion>
6
7     <groupId>org.example</groupId>
8     <artifactId>Exercise-3</artifactId>
9     <version>1.0-SNAPSHOT</version>
10
11     <properties>
12         <maven.compiler.source>25</maven.compiler.source>
13         <maven.compiler.target>25</maven.compiler.target>
14         <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
15     </properties>
16
17 </project>
```

Assertions.java:



```
1  import org.junit.jupiter.api.Test;
2  import static org.junit.jupiter.api.Assertions.*;
3
4  public class AssertionsTest {
5
6      @Test
7      void testAssertions() {
8          assertEquals( expected: 5, actual: 2 + 3);
9          assertTrue( condition: 5 > 3);
10         assertFalse( condition: 5 < 3);
11         assertNull( actual: null);
12         assertNotNull(new Object());
13     }
14 }
```

Output Screenshot:



```
Run  Main x
C:\Users\lokak\.jdk\openjdk-25.0.1\bin\java.exe -javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2.2\lib\idea_rt.jar=50490* -D
Hello and welcome!
1 = 1
1 = 2
1 = 3
1 = 4
1 = 5
Process finished with exit code 0
```

Exercise 4: Arrange-Act-Assert (AAA) Pattern, Test Fixtures, Setup and Teardown Methods in JUnit:

Main.java:

```
Main.java × pom.xml (Exercise-4) © AAASetupTeardownTest.java
1 package org.example;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         System.out.println("Hello and welcome!");
7
8         for (int i = 1; i <= 5; i++) {
9             System.out.println("i = " + i);
10        }
11    }
12 }
```

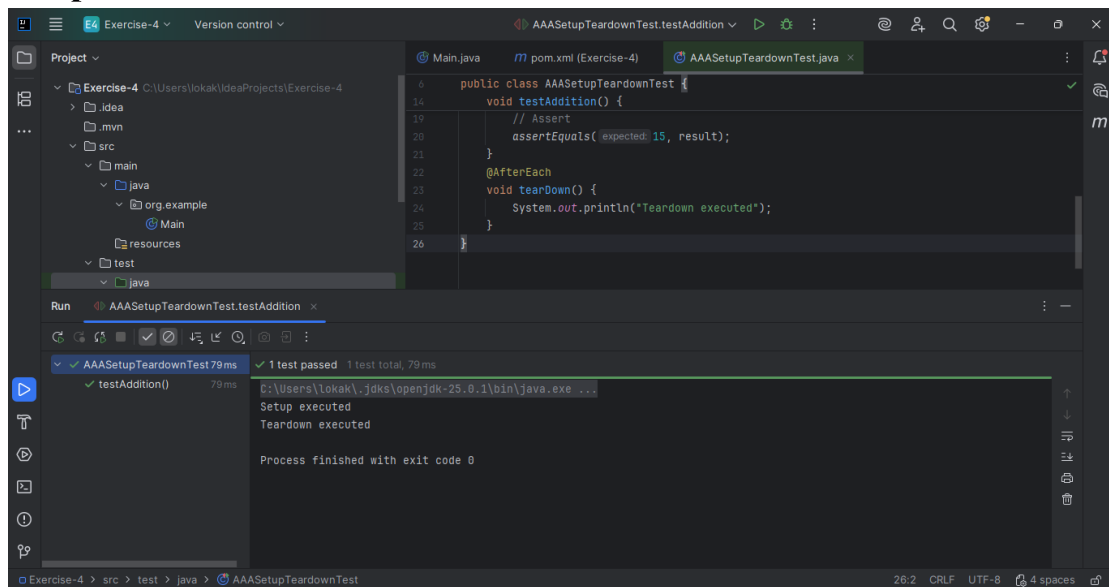
Pom.xml:

```
Main.java × pom.xml (Exercise-4) © AAASetupTeardownTest.java
1 <project xmlns="http://maven.apache.org/POM/4.0.0"
2     xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3     xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
4         https://maven.apache.org/xsd/maven-4.0.0.xsd">
5
6     <modelVersion>4.0.0</modelVersion>
7
8     <groupId>org.example</groupId>
9     <artifactId>Exercise-4</artifactId>
10    <version>1.0-SNAPSHOT</version>
11
12    <properties>
13        <maven.compiler.source>17</maven.compiler.source>
14        <maven.compiler.target>17</maven.compiler.target>
15    </properties>
16
17    <dependencies>
18        <dependency>
19            <groupId>org.junit.jupiter</groupId>
20            <artifactId>junit-jupiter</artifactId>
21            <version>5.10.2</version>
22            <scope>test</scope>
23        </dependency>
24    </dependencies>
25
26    <build>
```

AAASetupTeardownTest.java:

```
1  import org.junit.jupiter.api.AfterEach;
2  import org.junit.jupiter.api.BeforeEach;
3  import org.junit.jupiter.api.Test;
4  import static org.junit.jupiter.api.Assertions.assertEquals;
5
6  public class AAASetupTeardownTest {
7      private int number; 2 usages
8      @BeforeEach
9      void setUp() {
10         System.out.println("Setup executed");
11         number = 10;
12     }
13     @Test
14     void testAddition() {
15         // Arrange
16         int value = number;
17         // Act
18         int result = value + 5;
19         // Assert
20         assertEquals(expected: 15, result);
21     }
22     @AfterEach
23     void tearDown() {
24         System.out.println("Teardown executed");
25     }
26 }
```

Output Screenshot:



The screenshot shows the IDE interface with the `AAASetupTeardownTest.java` file open. The `testAddition()` method is selected, and the Run window is open, showing the test execution results. The output indicates that the test passed successfully, with the setup and teardown methods executed as expected.

```
Run AAASetupTeardownTest.testAddition
✓ AAASetupTeardownTest 79ms ✓ 1 test passed 1 test total, 79ms
  ✓ testAddition() 79ms
    C:\Users\lokak\.jdk\openjdk-25.0.1\bin\java.exe ...
    Setup executed
    Teardown executed
    Process finished with exit code 0
```